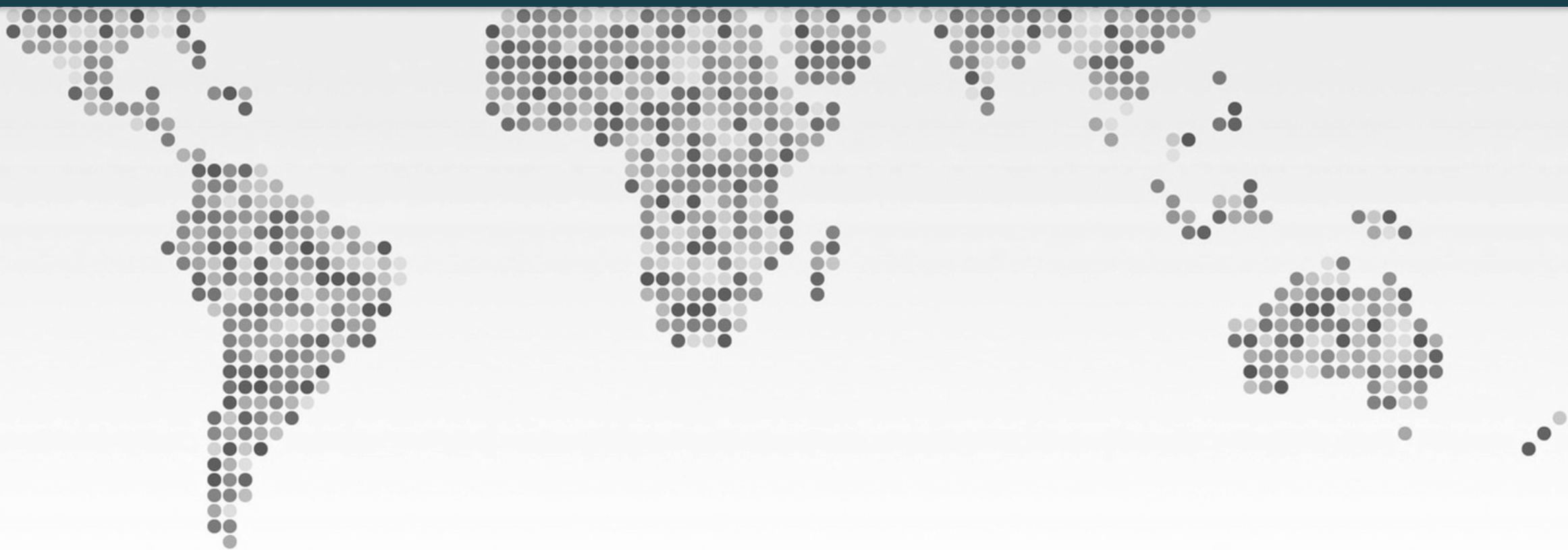


Software Engineering for Artificial Intelligence

Ethics and Fairness of Artificial Intelligence Solutions



Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: What It Is About

Machine learning systems need to live in a real environment and serve consumers. As such, their application needs to meet ethical regulations to be actually used.

While this has been always a need, *ethical machine learning* is becoming more and more important because of the growing complexity and reliance on those systems —> complexity is harmful, as machine/deep learning systems may take decisions hiding the nature of those decisions, possibly leading to unethical behavior.

Ethical machine learning is the union of many interrelated issues, like ethics, fairness, justice, discrimination, safety, privacy, security, transparency, and accountability. Each of these aspects is a deep and nuanced research topic!

Before diving into some of these aspects, let's first define two basic terms:

Legal

In accordance to societal laws. The systematic body of rules governing society. Something you get punishment for violation.

Ethical

Following moral principles of tradition, group, or individual. No legal binding, no enforcement beyond “shame”.

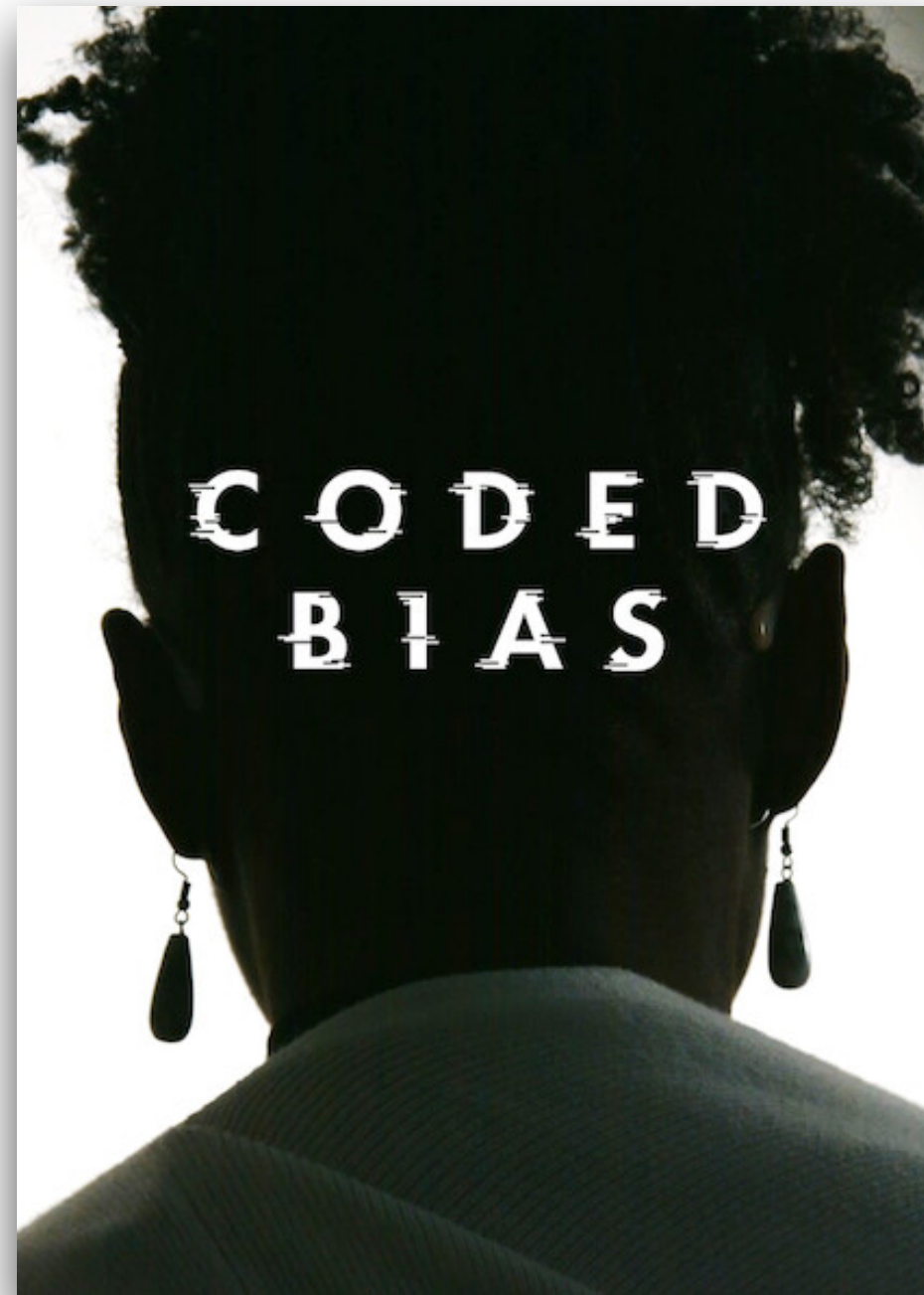
To say it better, ethics is a branch of philosophy and studies the standard human conduct in order to code rules used by professional organizations.

To some extent, ethics is more important than legal matters: high ethical standards may yield long-term benefits through image and loyalty.

Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: What It Is About

The real-world relevance of ethical machine learning is proved by the amount of movies and tv series on the matter.



and many others...

Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: What It Is About

Just some examples of ethical machine learning and its impact.

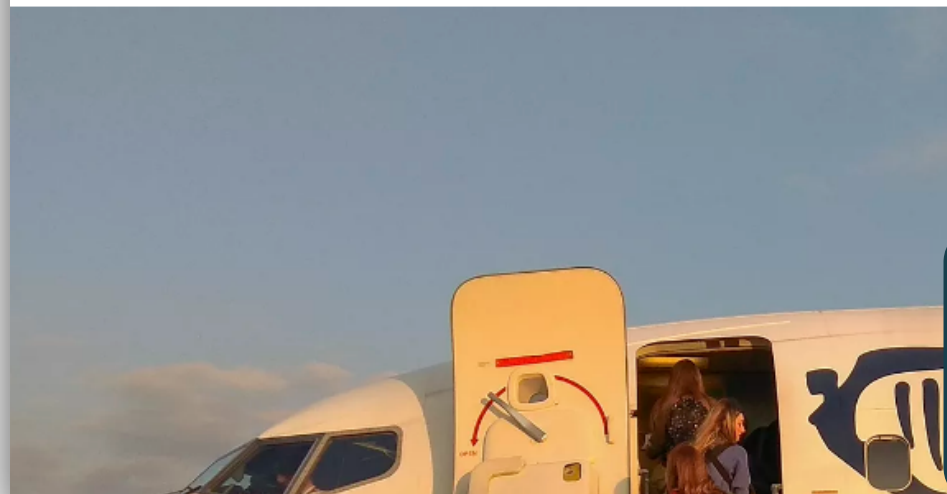
Some airlines may be using algorithms to split up families during flights

Your random airplane seat assignment might not be random at all.

By Aditi Shrikant | aditi@vox.com | Nov 27, 2018, 6:10pm EST



SHARE



This is just to say that the boundary between what is legal and what is ethical is often blurred and hard to draw.

A recent investigation by the UK's Civil Aviation Authority found that some airlines use algorithms to deliberately split up people traveling together to charge them an additional fee to change their seats.

In 2016, the US Congress passed the Families Flying Together Act, which would ensure that kids under the age of 13 would be seated next to their parent, but so far, the Federal Aviation Administration has not drafted new regulations to address the legislation.

Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: What It Is About

Just some examples of ethical machine learning and its impact.



Dr. Emily Slackerman Ackerman

@EmilyEAckerman

i (in a wheelchair) was just trapped *on* forbes ave by one of these robots, only days after their independent roll out. i can tell that as long as they continue to operate, they are going to be a major accessibility and safety issue. [thread]

[Traduci il Tweet](#)



Even in this case, the boundary is quite blurred. A IoT device is not supposed to threat accessibility/ safety, but in practice it may happen.

pittnews.com

Everything we know about the Starship food delivery robot
The white, 2-foot tall battery-powered delivery robots will roll on sidewalk with Oakland pedestrians starting sometime in

Machine learning systems do not only revolve around algorithms, but entire internet-of-things devices. What does this imply for society and how are we dealing with it?

Ethical Artificial Intelligence: What It Is About

Just some examples of ethical machine learning and its impact.

NO MERCY NO MALICE

Robinhood Has Gamified Online Trading Into an Addiction

Tech's obsession with addiction will hurt us all



Scott Galloway [Follow](#)

Jun 23 · 7 min read ★



Warning: This post contains a discussion of suicide.

Addiction is the inability to stop consuming a chemical activity although it's causing harm.

I engage with almost every substance or behavior associated with addiction: alcohol, drugs, coffee, porn,

Gamification is a recent trend that aims at embedding machine learning algorithms within games so that people get engaged more.

But machine learning may impact even differently on society, especially when used in trading activities.

Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: What It Is About

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healthline

SUBSCRIBE



stop hoarding and work with your ...
@jackyalcine

Google Photos, y'all fucked up. My fr
not a gorilla.



Skyscrapers



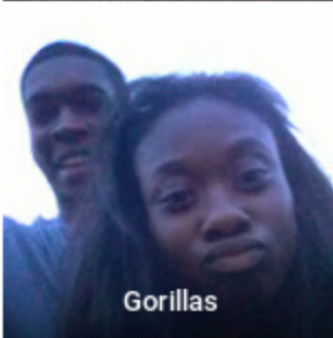
Airplanes



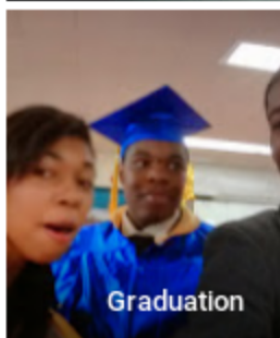
Cars



Bikes



Gorillas



Graduation

6:22 PM - 28 Jun 2015

3,352 Retweets 2,767 Likes



Kahuna

232

3.4K

2.8K

Artificial intelligence was asked to make a picture of Mother Teresa fighting against poverty.



Polarization

By [Jeff Horwitz](#) and [Deepa Seetharaman](#)
May 26, 2020 11:38 am ET

And even more... Mental health

Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: What It Is About

To regulate ethics in machine learning, the European Union has proposed the so-called, well-known **Artificial Intelligence Act**.

This represents the first law on AI by a major regulator anywhere - like the EU's GDPR in 2018, the EU AI Act could become a global standard.

AI-systems will be classified according to their risk levels:

Risk Level	Examples	Obligation
Unacceptable	Social scoring, manipulation of behavior, mass surveillance	Banned
High Risk	Biometrics, critical infrastructure, education, recruitment, credit scoring	Strict requirements on data, transparency, human oversight
Limited Risk	Chatbots, customer service bots	Must inform users they are interacting with AI
Minimal Risk	Spam filters, recommendation systems	Few or no obligations

Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: What It Is About

Just some examples of ethical machine learning and its impact.

Discrimination

Unequal treatment, unequal outcomes obtained by reinforcing patterns in predicting policing with feedback loops.

So, when designing machine learning solutions we cannot avoid ethical considerations. Many of the examples shown are ethically problematic, but some are perhaps legal. Are they intentional? Are they the cause of negligence? Were they really unforeseeable?

While we cannot know what happened, we should be aware of the existence of legally protected classes.

- Color (Civil Rights Act of 1964);
- Race (Civil Rights Act of 1964);
- Sex (Equal Pay Act of 1963; Civil Rights Act of 1964);
- Religion (Civil Rights Act of 1964);
- National origin (Civil Rights Act of 1964);
- Citizenship (Immigration Reform and Control Act);
- Age (Age Discrimination in Employment Act of 1967);
- Pregnancy (Pregnancy Discrimination Act);
- Familial status (Civil Rights Act of 1968);
- Disability status (Rehabilitation Act of 1973; Americans with Disabilities Act of 1990);
- Veteran status (Vietnam Era Veterans' Readjustment Assistance Act of 1974);
- Genetic information (Genetic Information Nondiscrimination Act).

FAIRNESS AND MACHINE LEARNING

Limitations and Opportunities

Solon Barocas, Moritz Hardt, Arvind Narayanan

<https://fairmlbook.org/pdf/fairmlbook.pdf>

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While we cannot know what happened, we should be aware of the existence of legally protected classes.

But also of regulated domains.

- Credit (Equal Credit Opportunity Act);
- Education (Civil Rights Act of 1964; Education Amendments of 1972);
- Employment (Civil Rights Act of 1964);
- Housing (Fair Housing Act);
- Public Accommodation (Civil Rights Act of 1964).

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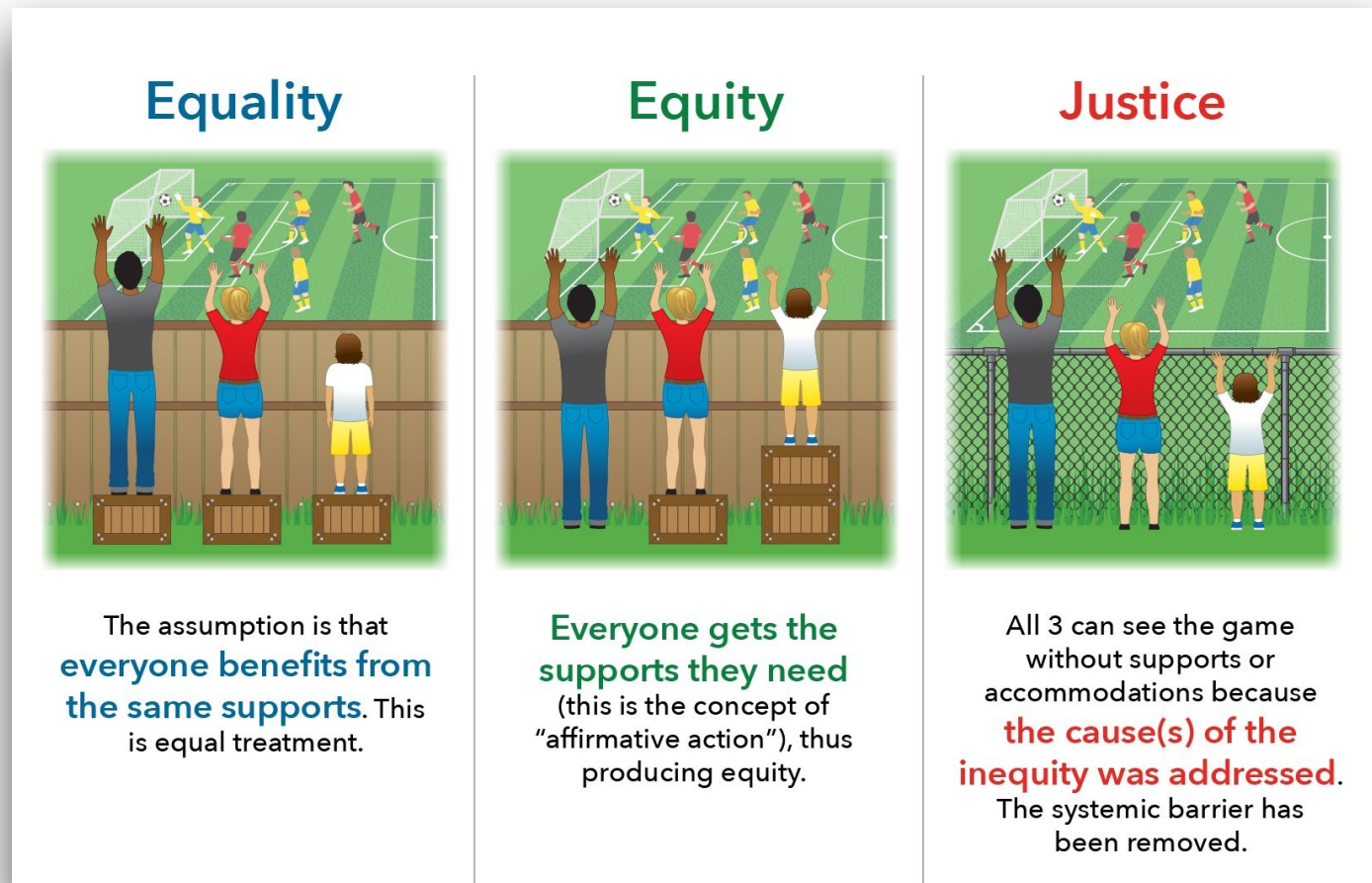
But also of the various levels of actions you may take.

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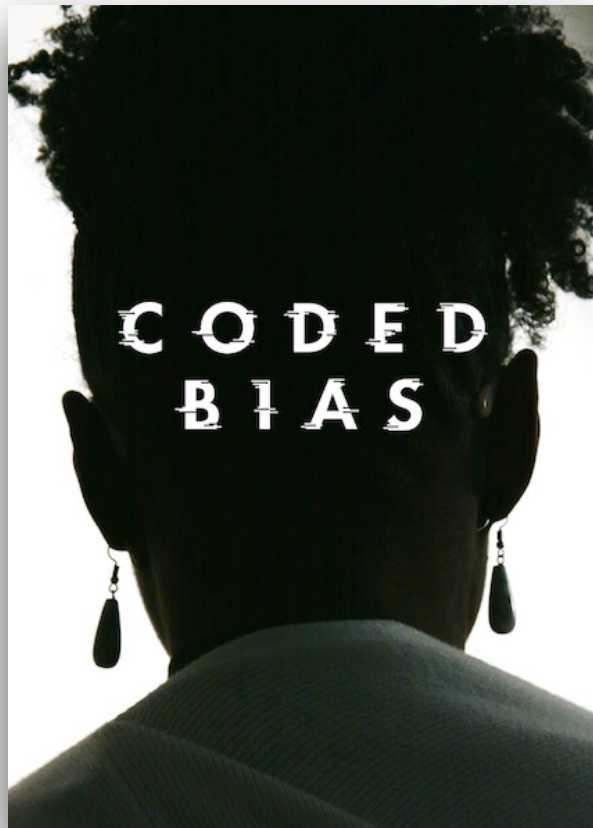
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And, sadly, independently from what you know and how careful you might be when designing ethical artificial intelligence solutions, problems might still arise.



“When you think of A.I., it’s forward-looking. But A.I. is based on data, and data is a reflection of our history.”

Artificial intelligence applications use previous data to make predictions or find patterns that will be used in the future, which may be sometimes problematic.

Consider the example of recruiting computer scientists: if computer science has been (BTW, wrongly) historically associated to men, how should an AI work?

Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: What It Is About

SOFTWARE MAINTENANCE AND EVOLUTION INTERMEZZO - DID YOU KNOW?

Computer Science has been often considered as a typical man activity: this is a huge historical forgery. *Margaret Hamilton* was likely the first software engineer. You might have heard about her already: she was the leader of the team involved in the NASA Apollo program that led to the landing of the first humans on the Moon.



<https://www.youtube.com/watch?v=ZbVOF0Uk5IU>



Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: What It Is About

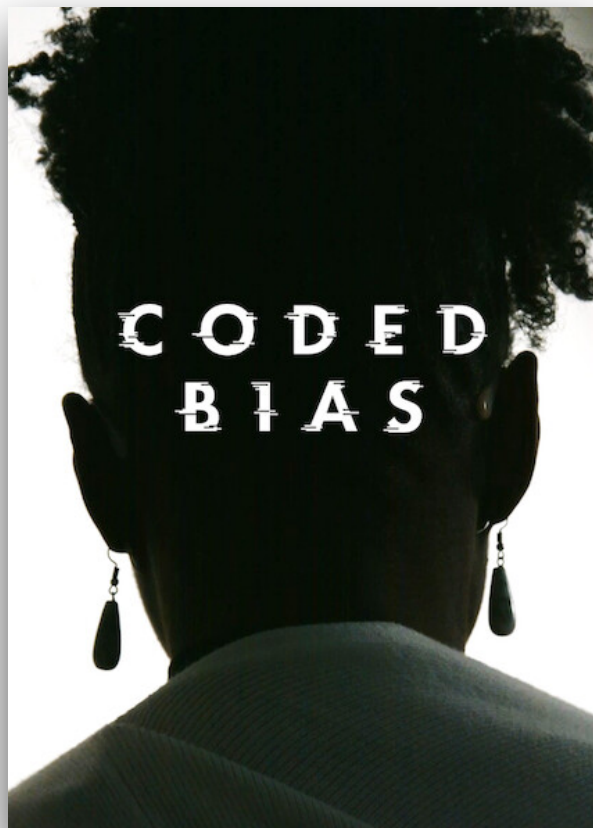
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“When you think of A.I., it’s forward-looking. But A.I. is based on data, and data is a reflection of our history.”

And always remember: Everyone is biased toward something! Our culture, our education, and others, influence us in many, many hidden things.

which may be sometimes problematic.

In this sense, ethical concerns are embedded in the project solution domain. Some issues arise by design (others may be even maliciously coded).

as associated to men, how should an AI work?

NB: This is not something that data preprocessing can handle or address!

Long story short: Voluntarily or not, data can perpetuate society’s existing race-, class- and gender-based inequities.

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In Italy, especially in the south, **expressing emotions openly** is often seen as a sign of authenticity and sincerity. People tend to **speak with passion**, use expressive gestures, and often interrupt each other—not out of rudeness, but as a sign of engagement in the conversation. In group settings, **personal opinions** are shared freely and often debated energetically, even in informal settings like a family lunch.



In contrast, Japanese culture values **harmony (wa)** and **social cohesion**. Conversations are often more **reserved**, and it's common to **avoid direct disagreement**. Silence can be meaningful, and expressing one's emotions too openly can be considered inappropriate. Instead of saying "no" outright, Japanese people might say "it might be difficult", which is a softer, indirect way of expressing disagreement.

Software Engineering for Artificial Intelligence: Ethics and Fairness

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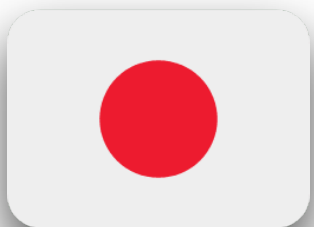
And, sadly, independently from what you know and how careful you might be when designing ethical artificial intelligence solutions, problems might still arise.



Suppose you are building a GAI agent that assists in **cross-cultural communication**, for example, a negotiation assistant or simply a chatbot.

If the training data mainly includes conversations, behaviors, or expectations from **one culture** (e.g., Western/Italian norms), the GAI agent may:

- Interpret **indirect speech** (typical in Japanese) as **evasive** or **uninformative**;
- Treat **direct, expressive speech** as the default ideal.



A GAI trained mainly on **Western customer service dialogues** might flag a Japanese speaker as “unsure” or “non-cooperative” simply because they use soft hedging like “*maybe*,” “*I’ll think about it*,” or pauses instead of a direct “yes”. But in their cultural context, this is **perfectly respectful and intentional**.

Software Engineering for Artificial Intelligence: Ethics and Fairness

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Fairness

Set of requirements, methods, and techniques to let an artificial intelligence solutions act “fairly”.

The first, foremost thing to say in this respect is concerned with the role of requirements engineering. The identification of the fairness concerns, possible feedback loops, and the mitigation/contingency plans are indeed part of the requirements engineering process.

In any case, it is important to remark that *not all discrimination is harmful!* For instance, gender discrimination in loan lending is illegal, but in medical diagnosis may be desirable.

In other terms, fairness and discrimination are *context-specific* concepts!

When referring to artificial intelligence algorithms, fairness is typically discussed in more technical terms with the use of the terms “*bias*” and “*discrimination*”.

The former refers to algorithmic issues, the latter to the distinguishing outcomes.

Software Engineering for Artificial Intelligence: Ethics and Fairness

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Selection bias

Disproportionate weight introduced by the selection of data for analysis;

Reporting bias

Disproportionate weight introduced by the suppression of information, e.g., when using surveys to collect data;

Bias of an estimator

Difference between the predictions made and the actual values.

Learning bias

Set of wrong assumptions that a learner uses to predict outputs.

Various sources lead to bias, e.g., limited features, sample size disparity, or other data collection problems. Also, AI is often designed by and for privileged/majority groups.

Also consider that predictions are made on correlations, not causation! They may depend on flawed statistics or deteriorated by feedback loops.

Ethical Artificial Intelligence: What It Is About

Fairness

Set of requirements, methods, and techniques to let an artificial intelligence solutions act “fairly”.

Given the contextual nature of software fairness, more than twenty different definitions have been provided so far, based on statistical, similarity, and casual reasoning terms.

Some basic concepts should be introduced:

- (1) We refer to a *protected* or *sensitive attribute* as an attribute of the dataset for which a machine learner may produce some form of discrimination. For instance, attributes referring to race or gender might be considered protected.
- (2) We refer to a *protected* or *sensitive group* as a group of individuals who might assume a value of a protected attribute that may lead to discrimination. For instance, in the known problem of recruitment, black people or women might be considered as protected groups.
- (3) We refer to the *real value of classification* as the category to which an individual belongs to and which is computed on the basis of her/his features.
- (4) We refer to the *prediction probability* as the conditioned probability that an individual is classified as belonging to a certain category.
- (5) We refer to the *prediction* as the decision taken by an algorithm with respect to the category to assign to a certain individual.

Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: What It Is About

Fairness

Set of requirements, methods, and techniques to let an artificial intelligence solutions act “fairly”.

Based on these concepts, we can reason on the implementation of the fairness concept in pure statistical terms.

Group fairness, a.k.a. Statistical parity

According to this definition, a binary machine learner is fair if individuals of a protected group G - defined based on a protected attribute p - are associated to the positive class with the same probability of assigning the individuals of non-protected groups.

Predictive parity, a.k.a. Outcome test

According to this definition, a binary machine learner is fair if both protected and non-protected groups have the same probability of assuming the real value of classification, i.e., the probability of individuals to be classified as true positive/negative is the same.

False positives Error rate balance, a.k.a. Predictive equality

According to this definition, a binary machine learner is fair if the probability of individuals to be associated to the positive class even though the real value of classification is opposite is the same for both protected and non-protected groups, i.e., the probability of being a false positive is the same.

These are just some of the definitions - many others come as we speak! In any case, it may be a good idea that of combining multiple definitions to have a better fairness idea.

Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: What It Is About

Fairness

Set of requirements, methods, and techniques to let an artificial intelligence solutions act “fairly”.

The statistical definitions suffer from a key issue: all are based on the concepts of *protected groups*, i.e., they all define fairness solely based on the protected attributes without considering other implementation constraints.

This treatment might potentially hide unfairness. Let's consider the case of statistical parity in the problem of recruitment: the same fraction of male and female applicants may be assigned a positive score, yet male applicants may still be chosen at random while female applicants may only be those that have the most savings —> without considering the whole set of attributes, it is likely to fall into trap and consider unfairness fair!

Casual discrimination

According to this definition, a binary machine learner is fair if it produces the same classification for any two subjects with the exact same attributes X . For instance, this implies that a male and female applicants who otherwise have the same attributes X will either both be hired or not.

Fairness through unawareness

According to this definition, a binary machine learner is fair if no sensitive attributes are explicitly used in the decision-making process.

The similarity-based definitions are therefore more holistic than the statistical ones.

Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: What It Is About

Fairness

Set of requirements, methods, and techniques to let an artificial intelligence solutions act “fairly”.

Causal reasoning definitions assume a given causal graph: a directed, acyclic graph with nodes representing attributes of an applicant and edges representing relationships between the attributes.

In theory, the relations between attributes and their influence on outcome are captured by a set of structural equations which are further used to provide methods to estimate effects of sensitive attributes and build algorithms that ensure a tolerable level of discrimination due to these attributes.

In practice, defining these relations is hard or even impossible. Think at a deep learner configured with multiple hidden layers: how can you describe cause-effect relations?

Anyway, provided a causal graph of this type, you may define fairness in more precise manners, such as:

Counterfactual fairness

According to this definition, a binary machine learner is fair if its predictions do not causally depend on any protected attributes.

So, depending on the definition of fairness, we might end up considering an artificial intelligence solution as fair or unfair.

Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: What It Is About (in Practice)

Fairness

Set of requirements, methods, and techniques to let an artificial intelligence solutions act “fairly”.

But this is not just a matter of definitions. Most of the established metrics assume availability of actual, verified outcomes. These outcomes are available for the training data, yet it is unclear if the real classified data always conforms to the same distribution. In addition, certain datasets have more protected attributes. Assessing fairness is extremely hard for one protected attribute at the time, consider two or more of them... To have a better idea of how hard is to assess fairness, a small exercise/reading for you.

Abstract: This dataset classifies people described by a set of attributes as good or bad credit risks. Comes in two formats (one all numeric). Also comes with a cost matrix

Data Set Characteristics:	Multivariate	Number of Instances:	1000	Area:	Financial
Attribute Characteristics:	Categorical, Integer	Number of Attributes:	20	Date Donated	1994-11-17
Associated Tasks:	Classification	Missing Values?	N/A	Number of Web Hits:	795669

[https://archive.ics.uci.edu/ml/datasets/statlog+\(german+credit+data\)](https://archive.ics.uci.edu/ml/datasets/statlog+(german+credit+data))

This dataset contains 20 features: from credit amount to savings accounts and bonds, from personal status and gender to the employment length, the dataset can be used to predict credit score, i.e., whether a person will get credit.

Based on the definition of fairness you choose, you might end up considering a classifier to act fairly or not. Read more here: <https://fairware.cs.umass.edu/papers/Verma.pdf>.

Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: Dealing With Fairness

And so, how can we deal with software fairness using software engineering methods?

Pre-processing

Bias mitigation algorithms applied to training data, namely **before** the actual training process

Examples of pre-processing algorithms refer to fair data balancing, fair data transformation, and fairness-driven data preprocessing.

To have an idea of what these algorithms are about, let's consider the case of FairSMOTE.

In-processing

Post-processing

Bias in Machine Learning Software: Why? How? What to Do?

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Raleigh, USA

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North Carolina State University
Raleigh, USA

ABSTRACT

Increasingly, software is making autonomous decisions in case of criminal sentencing, approving credit cards, hiring employees, and so on. Some of these decisions show bias and adversely affect certain social groups (e.g. those defined by sex, race, age, marital status). Many prior works on bias mitigation take the following form: change the data or learners in multiple ways, then see if any of that improves fairness. Perhaps a better approach is to postulate root causes of bias and then applying some resolution strategy.

This paper checks if the root causes of bias are the prior decisions about (a) what data was selected and (b) the labels assigned to those examples. Our Fair-SMOTE algorithm removes biased labels; and rebalances internal distributions so that, based on sensitive attribute, examples are equal in positive and negative classes. On testing, this method was just as effective at reducing bias as prior approaches. Further, models generated via Fair-SMOTE achieve higher performance (measured in terms of recall and F1) than other state-of-the-art fairness improvement algorithms.

To the best of our knowledge, measured in terms of number of analyzed learners and datasets, this study is one of the largest studies on bias mitigation yet presented in the literature.

CCS CONCEPTS

• Software and its engineering → Software creation and management; • Computing methodologies → Machine learning.

KEYWORDS

Software Fairness, Fairness Metrics, Bias Mitigation

ACM Reference Format:

Joymallya Chakraborty, Suvodeep Majumder, and Tim Menzies. 2021. Bias in Machine Learning Software: Why? How? What to Do?. In *Proceedings of the 29th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE '21)*, August 23–28, 2021, Athens, Greece. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3468264.3468537>

1 INTRODUCTION

It is the ethical duty of software researchers and engineers to produce quality software that makes fair decisions, especially for high-stake software that makes important decisions about human lives.

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ESEC/FSE '21, August 23–28, 2021, Athens, Greece
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ACM ISBN 978-1-4503-8562-6/21/08...\$15.00
<https://doi.org/10.1145/3468264.3468537>

Sadly, there are too many examples of machine learning software exhibiting unfair/biased behavior based on some protected attributes like sex, race, age, marital status. For example:

- Amazon had to scrap an automated recruiting tool as it was found to be biased against women [1].
- A widely used face recognition software was found to be biased against dark-skinned women [2].
- Google Translate, the most popular translation engine in the world, shows gender bias. "She is an engineer, He is a nurse" is translated into Turkish and then again into English becomes "He is an engineer, She is a nurse" [3].

Prior works on managing fairness in machine learning software have tried mitigating bias by exploring a very wide space of control parameters for machine learners. For example, Johnson et al. [4] executed some (very long) grid search that looped over the set of control parameters of a learner to find settings that reduced bias. Another approach, published in FSE'20 by Chakraborty et al. [5] used stochastic sampling to explore the space of control options (using some incremental feedback operator to adjust where to search next). While these approaches were certainly useful in the domains tested by Johnson and Chakraborty et al., these methods are "dumb" in a way because they do not take advantage of domain knowledge. This is a problem since, as shown by this paper, such domain knowledge can lead to a more direct (and faster and more effective) bias mitigation strategy.

The insight we offer here is that the root causes of bias might be the *prior decisions that generated the training data*. Those prior decisions affect (a) what data was collected and (b) the labels assigned to those examples. Hence, to fix bias, it might suffice to apply mutators to the training data in order to

(A) remove biased labels; and (B) rebalance internal distributions such that they are equal based on class and sensitive attributes.

This "Fair-SMOTE" tactic uses *situation testing* [6, 7] to find biased labels. Also, it balances frequencies of sensitive attributes and class labels (whereas the older SMOTE algorithm [8] just balances the class labels). We recommend Fair-SMOTE since:

- Fair-SMOTE is *faster* (220%) than prior methods such as Chakraborty et al.'s [5] method.
- Models generated via Fair-SMOTE are more *effective* (measured in terms of recall and F1) than Chakraborty et al. [5] and another state-of-the-art bias mitigation algorithm [9].

Overall, this paper makes the following contributions:

- We demonstrate two main reasons for the training data being biased - a) data imbalance, b) improper data label.
- We combine finding bias, explaining bias, and removing bias.
- We show that traditional class balancing techniques damage the fairness of the model.

Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: Dealing With Fairness

And so, how can we deal with software fairness using software engineering methods?

Pre-processing

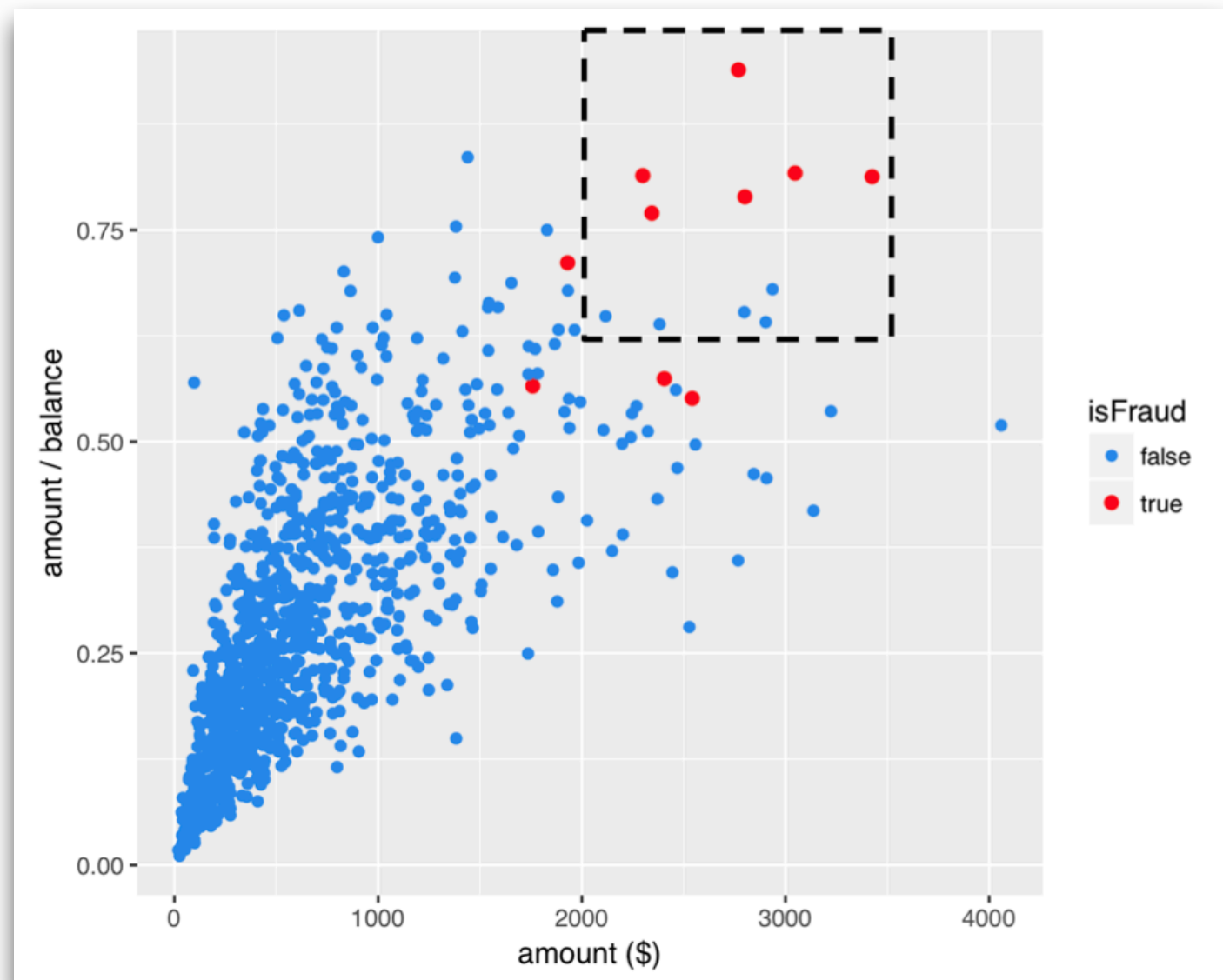
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To have an idea of what these algorithms are about, let's consider the case of FairSMOTE.

In-processing

Post-processing



Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: Dealing With Fairness

And so, how can we deal with software fairness using software engineering methods?

Pre-processing

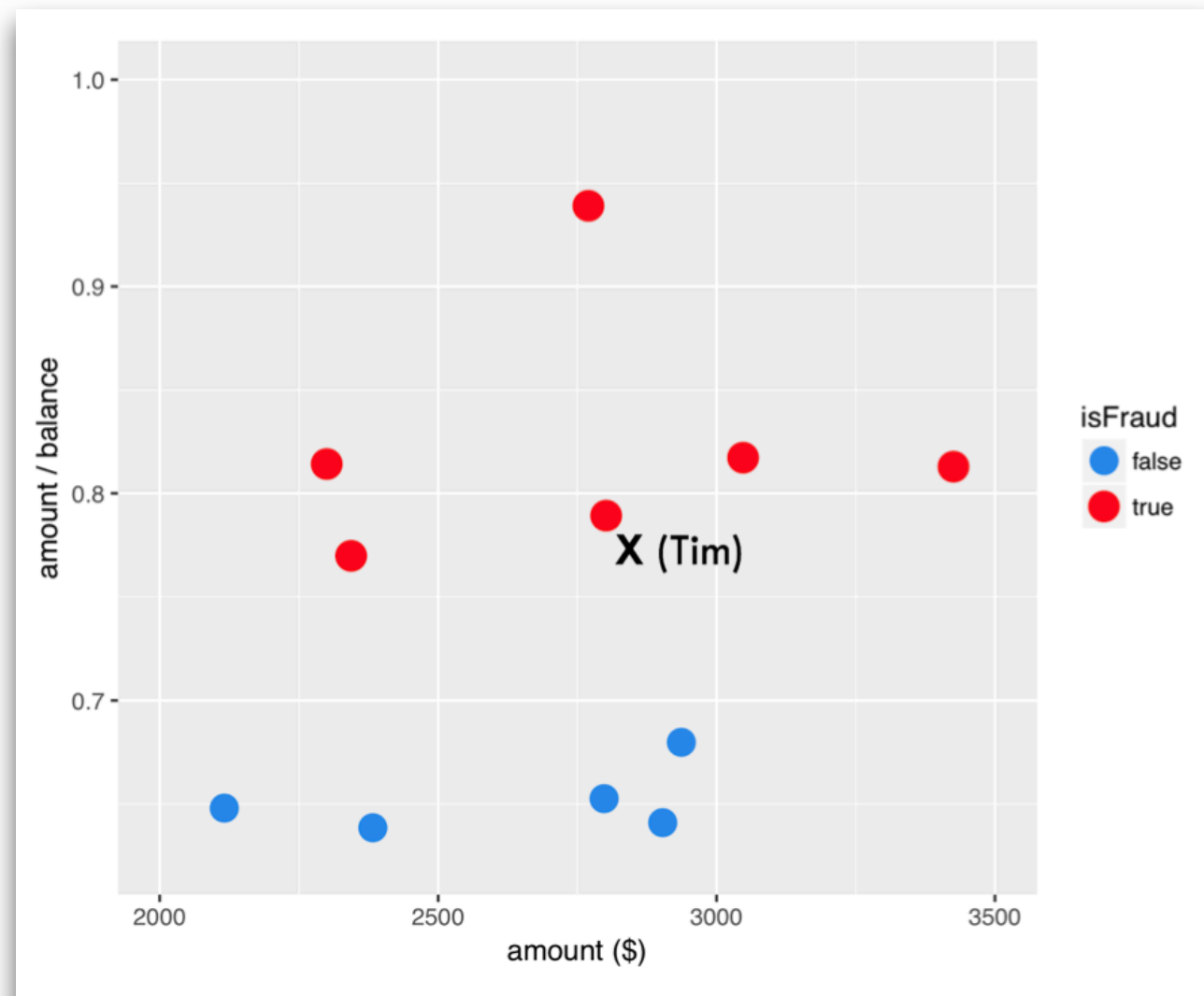
Bias mitigation algorithms applied to training data, namely **before** the actual training process

Examples of pre-processing algorithms refer to fair data balancing, fair data transformation, and fairness-driven data preprocessing.

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Select a random instance X of the minority class.

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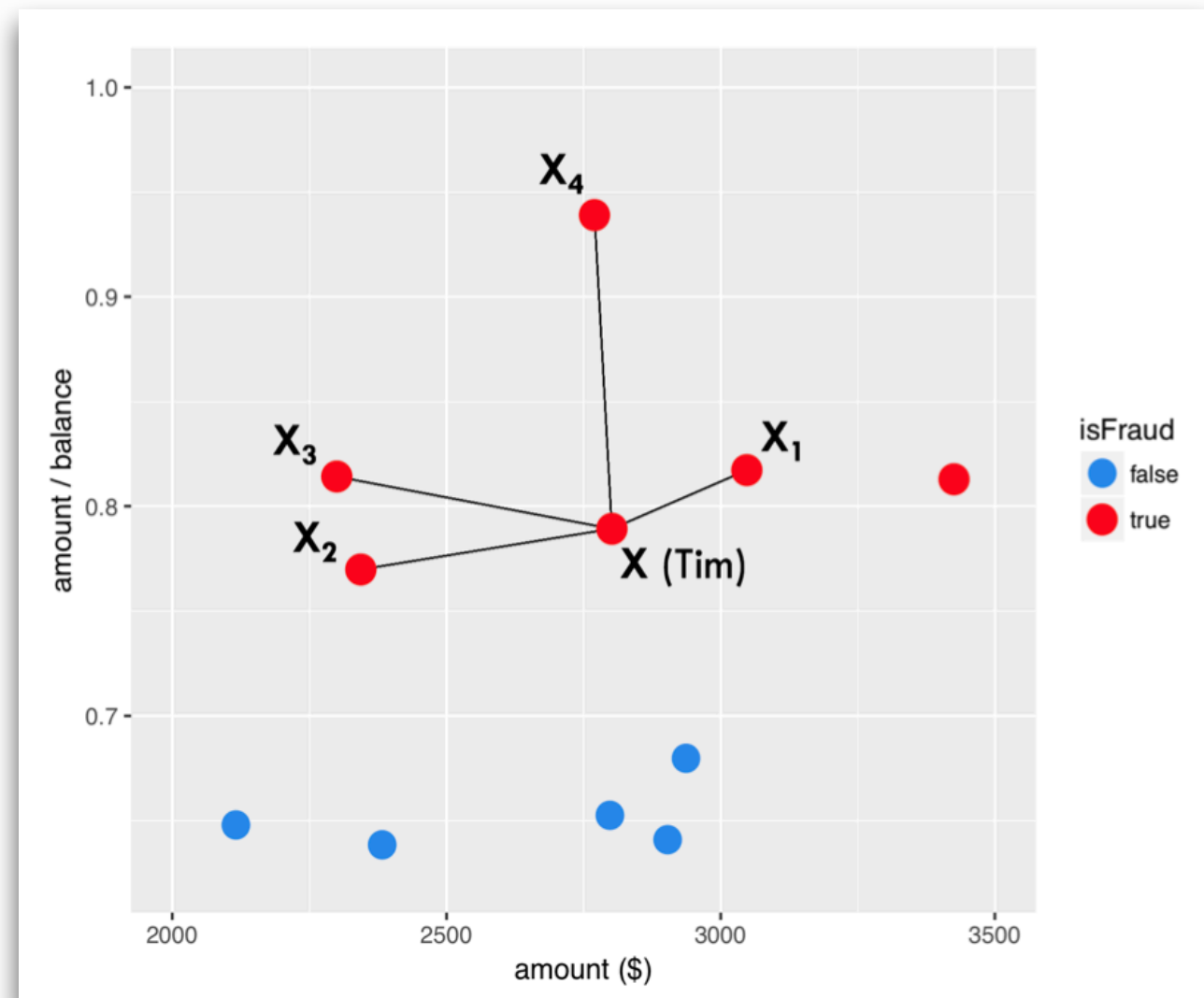
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Find the k -nearest neighbors of X , e.g., 4.

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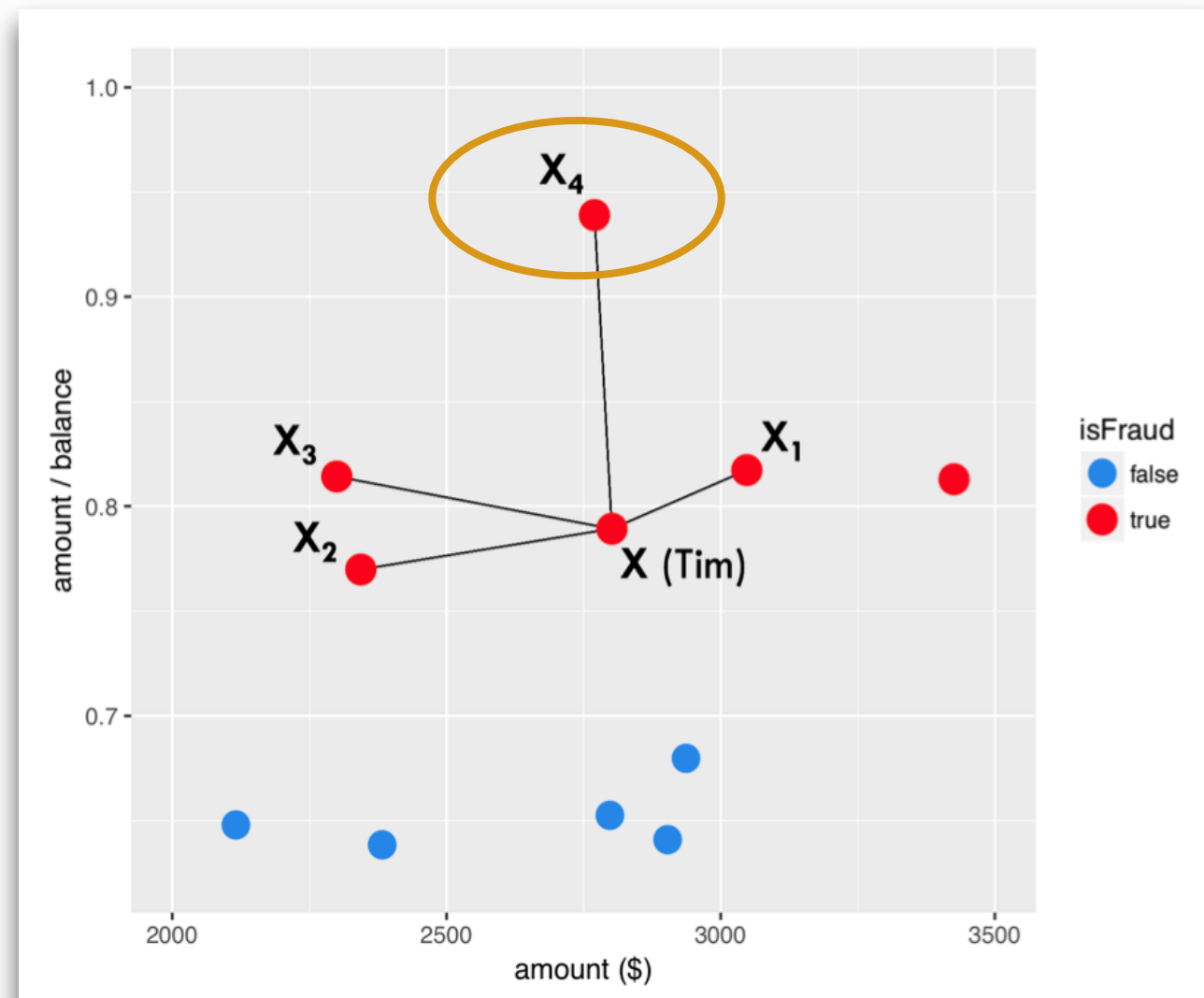
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Randomly choice of the the k -nearest neighbors of X .

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Post-processing

X_amount	X_ratio
2800	0.79

X4_amount	X4_ratio
2770	0.94

Choose random number in [0;1], e.g., 0.6.
Scaling of the data based on this number.

$2.800 + 0.6 * (2770 - 2800)$ ↓ $0.79 + 0.6 * (0.94 - 0.79)$ ↓

Xnew_amount	Xnew_ratio
2782	0.88

Generate a synthetic instance and repeat for all neighbors.

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Xnew_amount	Xnew_ratio
2782	0.88

Generate a synthetic instance and repeat for all neighbors.

Given this algorithm, how would you make SMOTE fair?

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In-processing

Post-processing

Traditional **SMOTE** balances only the **class distribution** (e.g., more positive examples). However, it does not consider **protected attribute distributions** and can actually worsen unfairness.

To account for this limitation, FairSMOTE partitions the training data into four subgroups (assuming binary class + binary protected attribute):

Class	Group	Example
Positive	Privileged	High income, male
Positive	Unprivileged	High income, female
Negative	Privileged	Low income, male
Negative	Unprivileged	Low income, female

Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: Dealing With Fairness

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To account for this limitation, FairSMOTE partitions the training data into four subgroups (assuming binary class + binary protected attribute).

Afterwards, it oversamples all subgroups except the one with the most data points.

In other terms, it introduces a layer where it selects the instances that should be oversampled according to a protected attribute.

The ultimate goal is to reach **equal representation** for each class per group combination.

Post-processing

Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: Dealing With Fairness

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In-processing

Bias mitigation algorithms applied to the **model itself** during the training phase

Examples of in-processing algorithms refer to fairness trade-offs mitigation, Generative Adversarial Training, Fair Graph Embedding Training

To have an idea of what these algorithms are about, let's consider the case of MAAT.

Post-processing

MAAT: A Novel Ensemble Approach to Addressing Fairness and Performance Bugs for Machine Learning Software

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ABSTRACT

Machine Learning (ML) software can lead to unfair and unethical decisions, making software fairness bugs an increasingly significant concern for software engineers. However, addressing fairness bugs often comes at the cost of introducing more ML performance (e.g., accuracy) bugs. In this paper, we propose MAAT, a novel ensemble approach to improving fairness-performance trade-off for ML software. Conventional ensemble methods combine different models with identical learning objectives. MAAT, instead, combines models optimized for different objectives: fairness and ML performance. We conduct an extensive evaluation of MAAT with 5 state-of-the-art methods, 9 software decision tasks, and 15 fairness-performance measurements. The results show that MAAT significantly outperforms the state-of-the-art. In particular, MAAT beats the trade-off baseline constructed by a recent benchmarking tool in 92.2% of the overall cases evaluated, 12.2 percentage points more than the best technique currently available. Moreover, the superiority of MAAT over the state-of-the-art holds on all the tasks and measurements that we study. We have made publicly available the code and data of this work to allow for future replication and extension.

CCS CONCEPTS

• Software and its engineering → Software creation and management; • Computing methodologies → Machine learning.

KEYWORDS

Software fairness, bias mitigation, fairness-performance trade-off, ensemble learning, machine learning software

ACM Reference Format:

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the Foundations of Software Engineering (ESEC/FSE '22), November 14–18, 2022, Singapore, Singapore. ACM, New York, NY, USA, 13 pages. <https://doi.org/10.1145/3540250.3549093>

1 INTRODUCTION

Software systems have been widely adopted to make decisions in social-critical, human-related tasks, including credit assessment [1], disease detection [55], criminal justice [4], and hiring [31]. The wide adoption of such decision-making software raises concerns about software fairness bugs (i.e., unfair software decisions) [27]. These bugs have been frequently reported, related to protected attributes such as race [5, 9] and sex [7, 17]. They may particularly disadvantage minorities and protected groups, resulting in unethical and unacceptable consequences.

The issue in fairness has been studied in both Software Engineering (SE) [40] and Machine Learning (ML) [62] research communities since 2008. From the SE perspective, fairness is a non-functional software property, as such can be the subject of testing to find fairness bugs [79]. In recent years, the surging ML software (i.e., software that relies on ML to tackle decision problems), increases the prevalence of software fairness bugs [58, 67]. Such prevalence attracts increasing attention from the SE community, and researchers have called for actions by the SE research to tackle fairness bugs [27]. As a result, the SE literature has witnessed a large number of recent results on addressing fairness bugs (i.e., software bias¹ mitigation) [25, 26, 28, 29, 46, 47, 78].

Although bias mitigation methods aim to address fairness bugs, many theoretical and empirical studies [23, 34, 36, 47, 73] have revealed that the reduction of fairness bugs can come at the cost of introducing more performance (e.g., accuracy) bugs into ML software, causing decrease in ML performance. Therefore, fairness and ML performance can be conflicting goals in software development. Software engineers are often in a dilemma between the functional property (i.e., ML performance) and the non-functional property (i.e., fairness), known as “fairness-performance trade-off”.

We present MAAT, a fairness-performance ensemble approach, for improving ML fairness-performance trade-off. Conventional ensemble methods combine different ML models with identical learning objectives (e.g., accuracy). In this paper, we propose a novel ensemble method that combines ML models with different learning objectives (i.e., fairness and performance) to optimize

¹We treat “bias” and “unfairness” as synonyms, referring to the opposite of “fairness”.

Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: Dealing With Fairness

And so, how can we deal with software fairness using software engineering methods?

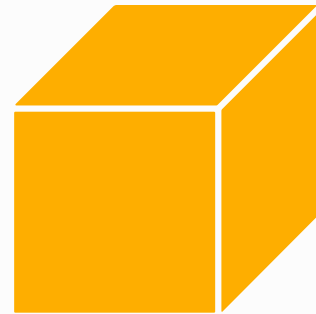
In-processing

Bias mitigation algorithms applied to the **model itself** during the training phase

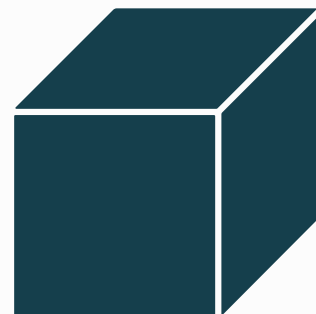
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To have an idea of what these algorithms are about, let's consider the case of MAAT.

Post-processing



In a standard scenario, you would train a model to maximize accuracy, meaning that you select an ML algorithm (e.g., Logistic Regression) and train it based on training data.



MAAT introduces an architectural solution where a second model, called FairModel, is trained. It uses the same or a different ML algorithm (e.g., a Random Forest) on a **debiased** training data.

A debiased dataset is a dataset that has been preprocessed to make the model less likely to learn discriminatory patterns, e.g., by applying FairSMOTE as a preprocessing step.

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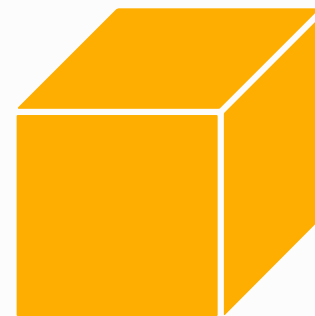
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Post-processing



performance output



fairness output

Ensemble layer

prediction

This ensemble layer can be customized, e.g., through Stacking or weighted average of the outcomes

Ethical Artificial Intelligence: Dealing With Fairness

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Post-processing

Bias mitigation algorithms applied **a-posteriori** to the predicted labels

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To have an idea of what these algorithms are about, let's consider the case of automated repairing of fairness issues.

Empirical Software Engineering (2024) 29:36
<https://doi.org/10.1007/s10664-023-10419-3>



Search-based Automatic Repair for Fairness and Accuracy in Decision-making Software

Max Hort¹ · Jie M. Zhang² · Federica Sarro³ · Mark Harman³

Accepted: 25 October 2023 / Published online: 3 January 2024
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Abstract

Decision-making software mainly based on Machine Learning (ML) may contain fairness issues (e.g., providing favourable treatment to certain people rather than others based on sensitive attributes such as gender or race). Various mitigation methods have been proposed to automatically repair fairness issues to achieve fairer ML software and help software engineers to create responsible software. However, existing bias mitigation methods trade accuracy for fairness (i.e., trade a reduction in accuracy for better fairness). In this paper, we present a novel search-based method for repairing ML-based decision making software to simultaneously increase both its fairness and accuracy. As far as we know, this is the first bias mitigation approach based on multi-objective search that aims to repair fairness issues without trading accuracy for binary classification methods. We apply our approach to two widely studied ML models in the software fairness literature (i.e., Logistic Regression and Decision Trees), and compare it with seven publicly available state-of-the-art bias mitigation methods by using three different fairness measurements. The results show that our approach successfully increases both accuracy and fairness for 61% of the cases studied, while the state-of-the-art always decrease accuracy when attempting to reduce bias. With our proposed approach, software engineers that previously were concerned with accuracy losses when considering

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Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: Dealing With Fairness

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Post-processing

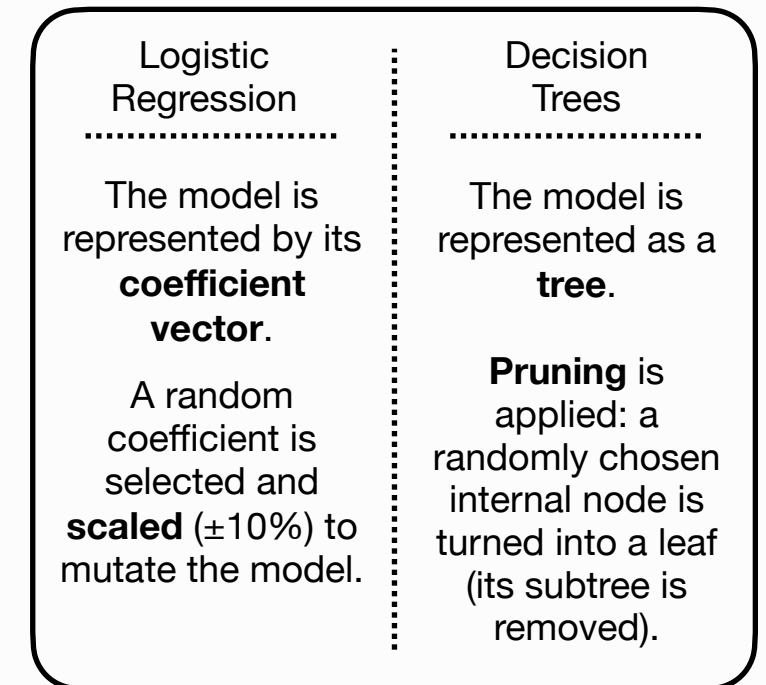
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trained model



mutation operators



Hill-Climbing Optimization

At each step, a mutation is applied. If the new model has higher accuracy and fairness, it's accepted. Otherwise, the mutation is reverted.



This is repeated until a **termination condition** is met. The best models are selected from the **Pareto front**, representing optimal trade-offs.

Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: Dealing With Fairness

And so, how can we deal with software fairness using software engineering methods?
Sure: *testing*!

Being the most important activity you might think at when thinking at software of any type, testing becomes relevant for fairness too.

Various methods are already available, even though the area is still to be considered as a scientific interest, meaning that new methods become available as we speak.

These methods attempt to operationalize the fairness definitions seen before in order to let them become testable. Let's see a couple of examples.

Anti-classification or Fairness through Blindness

Ignore/eliminate sensitive attributes from dataset, e.g., remove gender and race from a credit card scoring system.

While removing sensitive attributes is the easiest solution, remember that not all discriminations are harmful —> the blind decision to remove sensitive attributes is just stupid!

Yet, we might do a step ahead and create automated methods that verify how our artificial intelligence solution is balanced with respect to the decisions made on various instances of a dataset.

The testing of anti-classification revolves around the verification of an equation like this one:

$$\forall x . f(x[p \leftarrow 0]) = f(x[p \leftarrow 1])$$

The probability of an artificial intelligence to classify examples in two (or more) classes must be the same. The technique aims at generating random inputs and see how the system reacts in terms of this equation.

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$$\forall x. f(x[p \leftarrow 0]) = f(x[p \leftarrow 1])$$

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Verification

Given this formulation, you do not even need an oracle!
You can fully automate the testing procedure and just look at how the system behaves.

of an equation like this one:

$$\forall x. f(x[p \leftarrow 0]) = f(x[p \leftarrow 1])$$

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NB: This testing approach attempts to verify the predictive parity of examples —> the probability of individuals to be classified as true positive/negative is the same
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Software Engineering for Artificial Intelligence: Ethics and Fairness

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Classification parity

Classification error is equal across groups.

Testing for statistical parity, it aims at assessing the likelihood that positive predictions are the same across the groups.

The testing hypothesis is that the predictions must be independent from the sensitive attribute(s).

$$P[R = 1 | A = 0] = P[R = 1 | A = 1]$$

The equation aims at verifying the conditioned probability of a prediction being the same for a sensitive attribute A.

In more practical terms, this means separating validation data by protected attribute and separately measure the rate of positive predictions.

Be careful: this testing approach exercises the artificial intelligence solution for *equality*, but this might not be the case.

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The equation aims at verifying the conditioned probability of a

You might want to reformulate the testing objective by setting thresholds over specific groups. This makes the testing approach exercising for *equity*: on data of

$$P[R > t_0 | A = 0] = P[R > t_1 | A = 1]$$

Be careful: this testing approach exercises the artificial intelligence solution for *equality*, but this might not be the case.

Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: Achieving Fairness Criteria

Besides testing, we can achieve fairness during the entire learning process. In some cases, some additional steps are required.

Data acquisition

Testing starts from the beginning! Once you have acquired data, start testing for discrimination. Collect data if performance are poor on some groups.

This is, again, mainly something to tackle at requirements elicitation and analysis time. The analyses of the possible harms, other than the legacy and policy requirements, are indeed done at this time.

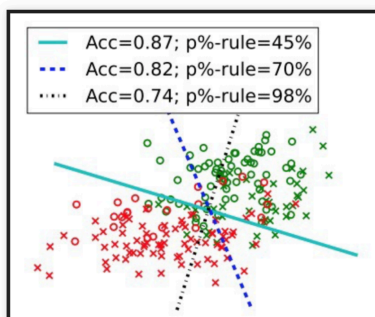
Take into account the guideline adopted by US Federal agencies (Department of Justice, Equal Employment Opportunity Commission, etc.) in 1978: the selection rate for a protected group is higher than 80% of the highest rate —> otherwise, the selection procedure is considered as “adverse impact”. If this is violated, the choice must be justified by business necessities.

Pre-processing

Clean the dataset to reduce correlation between the feature set and the sensitive attributes; in this way, you reduce the influence of sensitive data!

Training time constraints

When training your machine learner, impose additional parity constraints into the ML optimization process, e.g., as part of the loss function.



Remember that artificial intelligence solutions, like traditional software systems, work by means of trade-offs! Explicitly consider fairness as one of the objectives to reach. Fairness constraints will often lower the model accuracy for some group. That might and might be not fine, depending on the context. Yet, a reasoning of this type is needed!

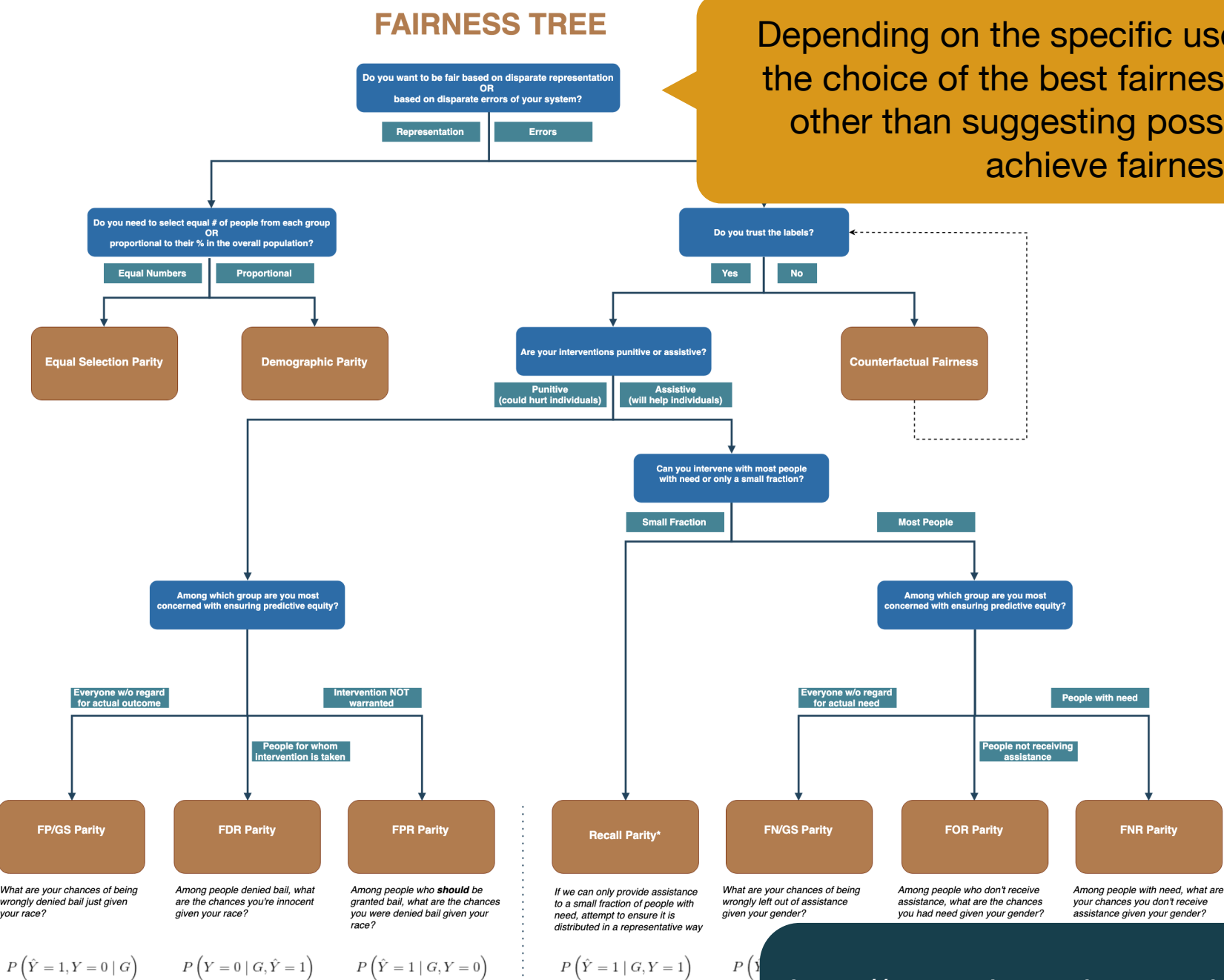
Post-processing

Adjust the learned model to be uncorrelated with sensitive attributes, for instance by testing multiple ways to set hyper-parameters.

Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: Achieving Fairness Criteria

Assessing fairness on-the-fly is possible. Aequitas is an open-source audit toolkit for data scientists and software engineers. It implements the so-called fairness tree:



Depending on the specific use case, the tree enables the choice of the best fairness definition to consider, other than suggesting possible actions to take to achieve fairness criteria.

<http://www.datasciencepublicpolicy.org/projects/aequitas/>

* Note: Focusing on recall in this case is equivalent to you may also want to reconsider the definition

Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: Achieving Fairness Criteria

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Data Source

- **Functional:** biases due to platform affordances and algorithms
- **Normative:** biases due to community norms
- **External:** biases due to phenomena outside social platforms
- **Non-individuals:** e.g., organizations, automated agents

Data Collection

- **Acquisition:** biases due to, e.g., API limits
- **Querying:** biases due to, e.g., query formulation
- **Filtering:** biases due to removal of data "deemed" irrelevant

Data Processing

- **Cleaning:** biases due to, e.g., default values
- **Enrichment:** biases from manual or automated annotations
- **Aggregation:** e.g., grouping, organizing, or structuring data

Data Analysis

- **Qualitative Analyses:** lack generalizability, interpret. biases
- **Descriptive Statistics:** confounding bias, obfuscated measurements
- **Prediction & Inferences:** data representation, perform. variations
- **Observational studies:** peer effects, select. bias, ignorability

Evaluation

- **Metrics:** e.g., reliability, lack of domain insights
- **Interpretation:** e.g., contextual validity, generalizability
- **Disclaimers:** e.g., lack of negative results and reproducibility

This because bias can be introduced at any stage of the data pipeline.

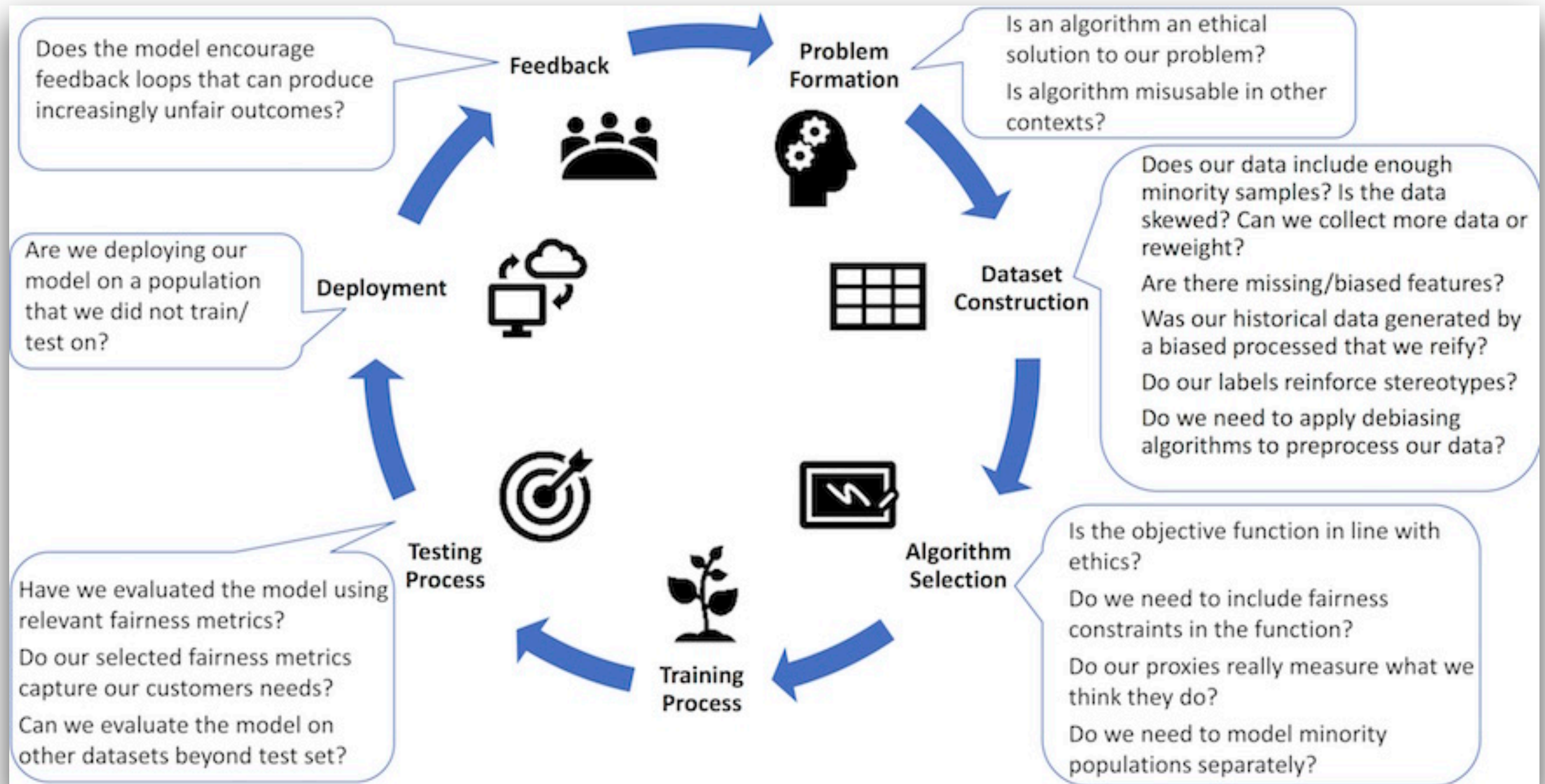
So, data acquisition becomes particularly relevant. You should answer the following questions:

- Address population bias: *Does the dataset reflect the demographics in the target population?*
- Address under- & over-representation issues: *Ensure sufficient amount of data for all groups to avoid being treated as "outliers" by ML, but also avoid over-representation of certain groups (e.g., remove historical data).*
- Data augmentation: Synthesize data for minority groups.
- Fairness-aware active learning: Apply procedures to collect more data for groups with highest error rates.

Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: Achieving Fairness Criteria

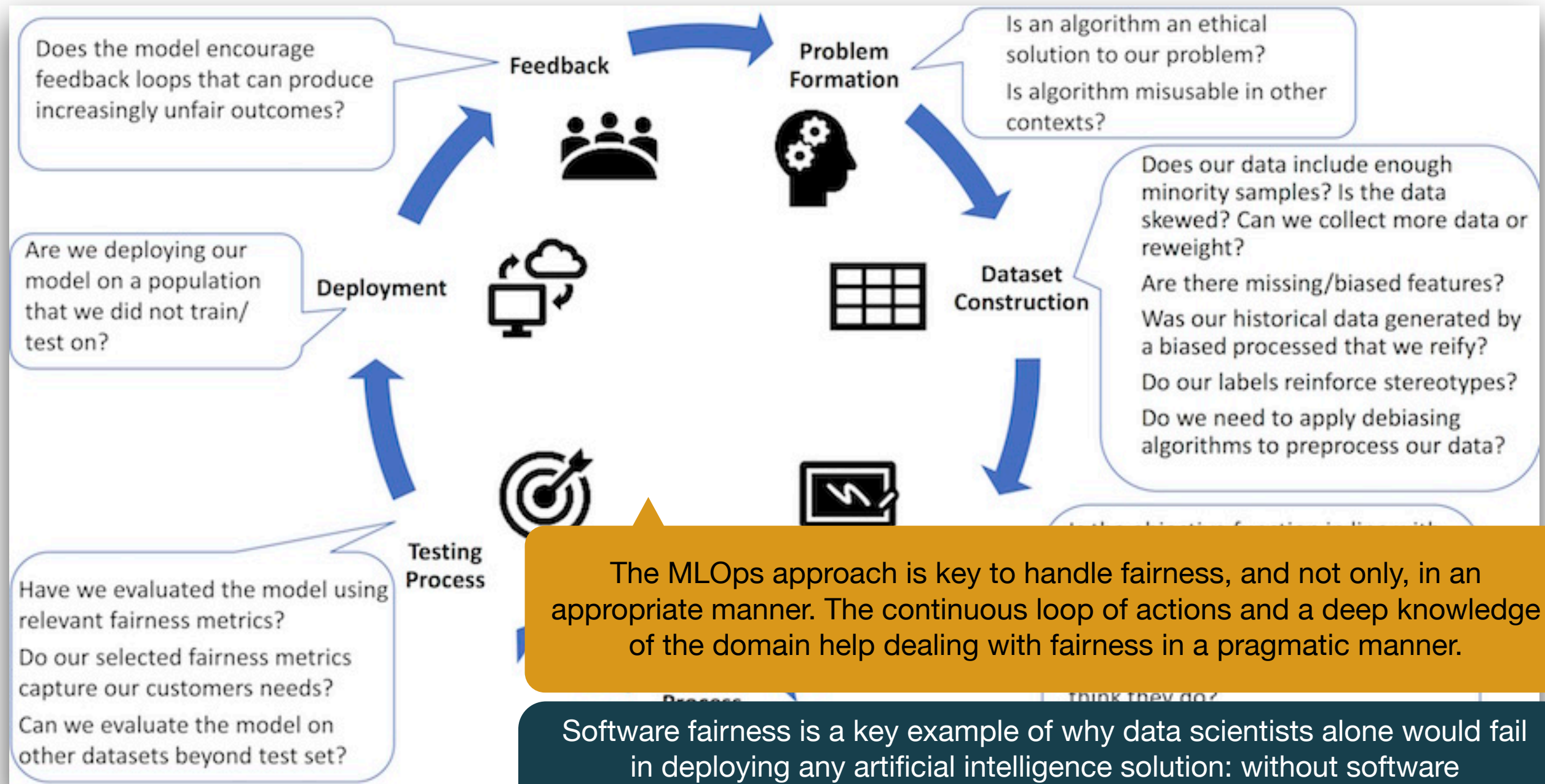
Assessing fairness throughout the lifecycle is instead a need. Nowadays, no artificial intelligence solution can be immune to fairness.



Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: Achieving Fairness Criteria

Assessing fairness throughout the lifecycle is instead a need. Nowadays, no artificial intelligence solution can be immune to fairness.



The MLOps approach is key to handle fairness, and not only, in an appropriate manner. The continuous loop of actions and a deep knowledge of the domain help dealing with fairness in a pragmatic manner.

Software fairness is a key example of why data scientists alone would fail in deploying any artificial intelligence solution: without software engineering, we would not have any artificial intelligence solutions readily available. Most of them would be enforced to be removed from the market!

Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: Key Challenges to Handle

So, to conclude, fairness practices can and should be applied throughout the software lifecycle, from requirements engineering till testing and monitoring.

A Catalog of Fairness-Aware Practices in Machine Learning Engineering

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ABSTRACT

Artificial Intelligence (AI)’s widespread adoption in decision-making processes, particularly with the introduction of AI-based assistants, raises concerns about ethics and fairness, particularly regarding the treatment of sensitive features and potential discrimination against underrepresented groups. The software engineering community has responded by developing fairness-oriented metrics, empirical studies, and mitigation approaches. However, there remains a gap in understanding and categorizing practices for engineering fairness throughout the development lifecycle of AI-based solutions. This paper presents a catalog of practices for addressing fairness derived from a systematic mapping study. The study identifies and categorizes 28 practices from existing literature, mapping them onto different stages of the development lifecycle. From this catalog, actionable items and implications for both researchers and practitioners in software engineering were extracted. This work aims to provide a comprehensive resource for integrating fairness considerations into the development and deployment of AI systems, enhancing their reliability, accountability, and credibility.

KEYWORDS

Fairness, AI Development Lifecycle, Software Engineering for AI.

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1 INTRODUCTION

Artificial Intelligence (AI) and Machine Learning (ML) have become integral to decision-making processes and task automation across

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various domains [1]. AI-enabled systems, software systems that include at least one component powered by AI algorithms [2], have demonstrated their effectiveness in numerous critical areas. However, ensuring the *quality* of these systems extends beyond their accuracy and efficiency—*fairness* is a fundamental property, particularly for AI systems that generate recommendations or influence human decisions [3]. The risk of bias in machine learning models remains a pressing challenge, as they can produce discriminatory outcomes against minorities due to biases present in historical training data. Several high-profile incidents have underscored the importance of fairness in AI systems [4]. For example, a Facebook vision model inappropriately labeled images of Black men as “primates,” Amazon’s algorithm unfairly lowered the sales rankings of books with LGBTQ+ themes, and authentication processes have exhibited biases based on race and gender. Moreover, with the recent introduction of AI-based assistants based on large language models (LLMs) [5, 6], such as Github’s CoPilot and ChatGPT, concerns regarding their ethical implications have emerged. Kotek et al. [7] tested AI-assistants to evaluate the fairness of their responses, revealing gender-related biased assumptions influenced by subjective opinions rather than factual evidence. Also, Truede et al. [8] found notable gender bias in software development tasks like issue assignment, highlighting the need for improved training in LLMs. These cases highlight how fairness is not just a desirable feature but an essential quality attribute for AI-enabled systems, particularly those interacting with or making recommendations to users.

Recognizing these ethical concerns, the research community increasingly treats ethics and fairness as critical non-functional properties of ML-enabled systems [9]. One approach to enhancing fairness has focused on *bias mitigation algorithms*, i.e., algorithmic solutions designed to address bias in AI models [10]. While these methods have demonstrated improvements in fairness, they have also been found to impact other non-functional requirements, such as accuracy or energy efficiency [11, 12]. Furthermore, recent research has highlighted that their adoption remains limited due to practical challenges in their application [13, 14]. To complement these efforts, researchers have also proposed various engineering practices to integrate fairness throughout the development pipeline, including fairness-aware training [15], hyper-parameter optimization [16], and automated fairness testing [17]. Instances of these practices have been explored across different types of AI-enabled systems, from balancing sensitive features in datasets for training

C1 - Requirements Elicitation & Analysis

1. Empirical methodologies for fair requirement elicitation and analysis
2. Multi-objective optimization for fairness constraints
3. Reverse engineering to elicit fairness requirement of a new system

C2 - Data Preparation

1. Data balancing techniques to respect fairness constraints
2. Data mining approaches to discover discrimination
3. Data & features transformation strategies under fairness constraints
4. Diversity data set selection for sensitive groups’ representations
5. Causal analysis approaches to identify discrimination dependencies in data
6. Measurement approaches to improve data fairness under multiple quality constraints
7. Multitask learning to maximize minority groups’ representativeness before training

C3 - Model Building

1. Ensemble learning strategies under different fairness definitions and constraints
2. Focused learning strategies to obtain discrimination-free outcomes
3. Fair regularization terms according to specific fairness metrics and constraints
4. Adversarial learning strategies to balance fairness in quality trade-offs of the model

C4 - Model Training & Testing

1. Fairness hyper-parameters tuning
2. Post-processing transformation to balance results among minority groups
3. Post-processing strategies to optimize fairness levels of the system
4. Fair test suites generation strategies
5. Mutation testing for unfairness causes the detection
6. Testing strategies based on correct prediction oracles

C5 - Model Verification & Validation

1. Validation strategies to detect discrimination according to different meanings of data
2. Features causal dependencies analysis to remove causes of discrimination
3. Model comparisons for fairness levels improvement
4. Definitions of fairness validation strategies among different definitions and metrics
5. Formal validation strategies to evaluate and improve fairness trade-offs

C6 - Model Maintenance & Evolution

1. Feature standardization to improve fairness
2. Model outcomes analysis to improve fairness
3. Multiple Datasets analysis to improve sensitive data representativeness

Paper available as part of the additional material!

Link to the related website: <https://sesafairnesspracticescatalog.github.io>

Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: Key Challenges to Handle

So, to conclude, fairness practices can and should be applied throughout the software lifecycle, from requirements engineering till testing and monitoring.

A Catalog of Fairness-Aware Practices from Developers' Perspective

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ABSTRACT

Artificial Intelligence (AI)'s widespread adoption in decision-making processes, particularly with the introduction of AI-based systems, raises concerns about ethics and fairness, particularly the treatment of sensitive features and potential discrimination against underrepresented groups. The software engineering community has responded by developing fairness-oriented metrics, studies, and mitigation approaches. However, there is a need for a better understanding and categorizing practices for engineering fairness throughout the development lifecycle of AI-based systems. This paper presents a catalog of practices for addressing fairness derived from a systematic mapping study. The study categorizes 28 practices from existing literature, mapped onto different stages of the development lifecycle. The catalog, actionable items and implications for both researchers and practitioners in software engineering were extracted and aims to provide a comprehensive resource for integrating fairness considerations into the development and deployment of AI-based systems, enhancing their reliability, accountability, and credibility.

KEYWORDS

Fairness, AI Development Lifecycle, Software Engineering

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1 INTRODUCTION

Artificial Intelligence (AI) and Machine Learning (ML) are becoming integral to decision-making processes and task automation in various domains.

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Fairness-Aware Practices from Developers' Perspective

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Abstract

Context. Machine Learning (ML) technologies have shown great potential when used without proper oversight, they can produce biased results, particularly for historically underrepresented groups. In recent years, the software engineering community has contributed to addressing the need for ethical machine learning by developing fairness-aware practices, e.g., fair data balancing or test management of fairness requirements throughout the software lifecycle. However, the validity of these practices, in terms of practical application, from the developers' perspective has not been investigated yet. **Objectives.** This paper assesses the developers' perspective of a set of 28 fairness-aware practices. **Methods.** We perform a survey study involving 155 practitioners in the development and maintenance of ML-enabled systems, and a clustering analysis to group fairness-aware practices based on their impact on bias mitigation, and effort required for their application. **Results.** We found that the most impactful practices are those applied at the early stages of the software lifecycle, and sometimes high, with a subsequent average application. **Conclusion.** The need for effort-aware automated approaches that ease the integration of fairness as well as recommendation systems that may suggest which practices throughout the software lifecycle.

Keywords: Software Engineering for Artificial Intelligence; Survey Studies; Empirical Software Engineering

1. Introduction

Artificial Intelligence (AI) and its branches, such as Machine Learning (ML), are becoming hot topics over the last decades, supporting humans in automating repetitive tasks to reduce effort and workload in various domains. Software systems that include at least one component powered by AI/ML are becoming increasingly common.

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Fairness on a Budget, Across the Board: A Cost-Effective Evaluation of Fairness-Aware Practices Across Contexts, Tasks, and Sensitive Attributes

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Abstract

Context. Machine Learning (ML) is widely used in critical domains like finance, healthcare, and criminal justice, where unfair predictions can lead to harmful outcomes. Although bias mitigation techniques have been developed by the Software Engineering (SE) community, their practical adoption is limited due to complexity and integration issues. As a simpler alternative, fairness-aware practices, namely conventional ML engineering techniques adapted to promote fairness, e.g., MinMax Scaling, which normalizes feature values to prevent attributes linked to sensitive groups from disproportionately influencing predictions, have recently been proposed, yet their actual impact is still unexplored. **Objective.** Building on our prior work that explored fairness-aware practices in different contexts, this paper extends the investigation through a large-scale empirical study assessing their effectiveness across diverse ML tasks, sensitive attributes, and application contexts. **Methods.** We conduct 5,940 experiments, evaluating fairness-aware practices from two perspectives: *contextual bias mitigation* and *cost-effectiveness*. Contextual evaluation examines fairness improvements across different ML models, sensitive attributes, and contexts. Cost-effectiveness analysis considers the trade-off between fairness gains and performance costs. **Results.** Findings reveal that the effectiveness of fairness-aware practices depends on context and configuration, while cost-effectiveness analysis highlights those that best balance ethical gains and efficiency. **Conclusion.** These insights guide practitioners in choosing fairness-enhancing practices with minimal performance impact, supporting ethical ML development.

Keywords: Software Engineering for Artificial Intelligence; Machine Learning Fairness Engineering; Cost-Effectiveness; Empirical Software Engineering.

1. Introduction

Artificial Intelligence (AI), with Machine Learning (ML) at its core, is rapidly integrating into daily life, automating decision-making processes [47, 71, 83]. However, its widespread adoption

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Software Engineering for Artificial Intelligence: Ethics and Fairness

Before leaving... What Should You Remember?

Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: What It Is About

Machine learning systems need to live in a real environment and serve consumers. As such, their application needs to meet ethical regulations to be actually used.

While this has been always a need, *ethical machine learning* is becoming more and more important because of the growing complexity and reliance on those systems —> complexity is harmful, as machine/deep learning systems may take decisions hiding the nature of those decisions, possibly leading to unethical behavior.

Ethical machine learning is the union of many interrelated issues, like ethics, fairness, justice, discrimination, safety, privacy, security, transparency, and accountability. Each of these aspects is a deep and nuanced research topic!

Before diving into some of these aspects, let's first define two basic terms:

Legal

In accordance to societal laws. The systematic body of rules governing society. Something you get punishment for violation.

Ethical

Following moral principles of tradition, group, or individual. No legal binding, no enforcement beyond "shame".

To say it better, ethics is a branch of philosophy and studies the standard human conduct in order to code rules used by professional organizations.

To some extent, ethics is more important than legal matters: high ethical standards may yield long-term benefits through image and loyalty.

Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: What It Is About

Just some examples of ethical machine learning and its impact.

Discrimination

Unequal treatment, unequal outcomes obtained by reinforcing patterns in predicting policing with feedback loops.

So, when designing machine learning solutions we cannot avoid ethical considerations. Many of the examples shown are ethically problematic, but some are perhaps legal. Are they intentional? Are they the cause of negligence? Were they really unforeseeable?

And, sadly, independently from what you know and how careful you might be when designing ethical artificial intelligence solutions, problems might still arise.



"When you think of A.I., it's forward-looking. But A.I. is based on data, and data is a reflection of our history."

Artificial intelligence applications use previous data to make predictions or find patterns that will be used in the future, which may be sometimes problematic.

In this sense, ethical concerns are embedded in the project solution domain. Some issues arise by design (others may be even maliciously coded).

NB: This is not something that data preprocessing can handle or address!

as associated to men, how should an AI work?

Long story short: Voluntarily or not, data can perpetuate society's existing race-, class- and gender-based inequities.

Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: What It Is About

Fairness

Set of requirements, methods, and techniques to let an artificial intelligence solutions act "fairly".

Based on these concepts, we can reason on the implementation of the fairness concept in pure statistical terms.

Group fairness, a.k.a. Statistical parity

According to this definition, a binary machine learner is fair if individuals of a protected group G - defined based on a protected attribute p - are associated to the positive class with the same probability of assigning the individuals of non-protected groups.

Predictive parity, a.k.a. Outcome test

According to this definition, a binary machine learner is fair if both protected and non-protected groups have the same probability of assuming the real value of classification, i.e., the probability of individuals to be classified as true positive/negative is the same.

False positives Error rate balance, a.k.a. Predictive equality

According to this definition, a binary machine learner is fair if the probability of individuals to be associated to the positive class even though the real value of classification is opposite is the same for both protected and non-protected groups, i.e., the probability of being a false positive is the same.

These are just some of the definitions - many others come as we speak! In any case, it may be a good idea that of combining multiple definitions to have a better fairness idea.

Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: Dealing With Fairness

And so, how can we deal with software fairness using software engineering methods? Sure: *testing*!

Being the most important activity you might think at when thinking at software of any type, testing becomes relevant for fairness too.

Various methods are already available, even though the area is still to be considered as a scientific interest, meaning that new methods become available as we speak.

These methods attempt to operationalize the fairness definitions seen before in order to let them become testable. Let's see a couple of examples.

Anti-classification or Fairness through Blindness

Ignore/eliminate sensitive attributes from dataset, e.g., remove gender and race from a credit card scoring system.

Classification parity

Classification error is equal across groups.

Testing for statistical parity, it aims at assessing the likelihood that positive predictions are the same across the groups.

The testing hypothesis is that the predictions must be independent from the sensitive attribute(s).

$$P[R = 1 | A = 0] = P[R = 1 | A = 1]$$

The equation aims at verifying the conditioned probability of a

You might want to reformulate the testing objective by setting thresholds over specific groups. This makes the testing approach exercising for equity:

$$P[R > \tau_0 | A = 0] = P[R > \tau_1 | A = 1]$$

Be careful: this testing approach exercises the artificial intelligence solution for *equality*, but this might not be the case.

Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: Achieving Fairness Criteria

Besides testing, we can achieve fairness during the entire learning process. In some cases, some additional steps are required.

Data Source

- Functional: biases due to platform affordances and algorithms
- Normative: biases due to community norms
- External: biases due to phenomena outside social platforms
- Non-individuals: e.g., organizations, automated agents

Data Collection

- Acquisition: biases due to, e.g., API limits
- Querying: biases due to, e.g., query formulation
- Filtering: biases due to removal of data "deemed" irrelevant

Data Processing

- Cleaning: biases due to, e.g., default values
- Enrichment: biases from manual or automated annotations
- Aggregation: e.g., grouping, organizing, or structuring data

Data Analysis

- Qualitative Analyses: lack generalizability, interpret. biases
- Descriptive Statistics: confounding bias, obfuscated measurements
- Prediction & Inferences: data representation, perform. variations
- Observational studies: peer effects, select. bias, ignorability

Evaluation

- Metrics: e.g., reliability, lack of domain insights
- Interpretation: e.g., contextual validity, generalizability
- Disclaimers: e.g., lack of negative results and reproducibility

This because bias can be introduced at any stage of the data pipeline.

So, data acquisition becomes particularly relevant. You should answer the following questions:

- Address population bias: Does the dataset reflect the demographics in the target population?
- Address under- & over-representation issues: Ensure sufficient amount of data for all groups to avoid being treated as "outliers" by ML, but also avoid over-representation of certain groups (e.g., remove historical data).
- Data augmentation: Synthesize data for minority groups.
- Fairness-aware active learning: Apply procedures to collect more data for groups with highest error rates.

Software Engineering for Artificial Intelligence: Ethics and Fairness

Ethical Artificial Intelligence: Achieving Fairness Criteria

Assessing fairness throughout the lifecycle is instead a need. Nowadays, no artificial intelligence solution can be immune to fairness.

Does the model encourage feedback loops that can produce increasingly unfair outcomes?

Are we deploying our model on a population that we did not train/test on?

Have we evaluated the model using relevant fairness metrics? Do our selected fairness metrics capture our customers needs? Can we evaluate the model on other datasets beyond test set?

Problem Formation

Is an algorithm an ethical solution to our problem? Is algorithm misusable in other contexts?

Dataset Construction

Does our data include enough minority samples? Is the data skewed? Can we collect more data or reweight?

Are there missing/biased features? Was our historical data generated by a biased processed that we rely? Do our labels reinforce stereotypes? Do we need to apply debiasing algorithms to preprocess our data?

Testing Process

The MLOps approach is key to handle fairness, and not only, in an appropriate manner. The continuous loop of actions and a deep knowledge of the domain help dealing with fairness in a pragmatic manner.

Software fairness is a key example of why data scientists alone would fail in deploying any artificial intelligence solution: without software engineering, we would not have any artificial intelligence solutions readily available. Most of them would be enforced to be removed from the market!

Software Engineering for Artificial Intelligence

Ethics and Fairness of Artificial Intelligence Solutions

